

Proposal 1: Heatwave Dataset – Rajasthan, India (2006–2025)

<https://www.kaggle.com/datasets/rupsarroy/heatwave-dataset-rajasthan-india-2006-2025>

This dataset covers nearly two decades (2006–2025) of heatwave-related climate records for Rajasthan, India. Heat is a growing public health and infrastructure risk in India, and this dataset lets us work with a real, high-stakes climate problem rather than a generic dataset. Our goal is to build a model that predicts whether a given day (or short period) will qualify as a heatwave day, based on temperature and related meteorological features. This is a binary classification problem, and depending on the exact columns available, we may also explore predicting maximum temperature as a regression task.

Preprocessing plan

- Handle missing/null values in temperature or humidity columns (median fill depending on gaps)
- Check for and handle outliers (sensor errors vs. genuine extreme heat)
- Scale/normalize temperature and humidity features
- Encode any categorical fields (season, month, region)
- Feature engineering: rolling averages, day-of-year, anomaly from historical mean

Proposal 2: Online Shoppers Purchasing Intention Dataset

<https://www.kaggle.com/datasets/imakash3011/online-shoppers-purchasing-intention-dataset>

This dataset contains 12,330 online shopping sessions, each represented as a feature vector describing browsing behavior over a one-year period (originally from the UCI Machine Learning Repository). Features include session-level metrics like number of administrative/informational/product-related pages visited, time spent on each page type, bounce rate, exit rate, page value, plus contextual fields like month, visitor type (new/returning), weekend flag, and operating system/browser info. The target variable, "Revenue," indicates whether the session ended in a purchase. It's a well-documented, real-world e-commerce dataset well above the 2,000-row minimum, with a natural binary classification target. We want to build a model that predicts purchase intent (Revenue: true/false) from session browsing behavior, for example, Shopify has something similar. This has a clear practical use in an e-commerce platform that could use this to flag high-intent sessions in real time and signal the owner of the e-commerce store to start promotions or deals.

Preprocessing plan

- Check for and handle any missing values
- Encode categorical fields (Month, VisitorType, Weekend) via one-hot encoding
- Scale numerical features (page counts, durations, etc.)

- Address class imbalance in the Revenue target using SMOTE or similar oversampling
- Feature engineering: possibly combine page-count and duration features into engagement ratios